THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Powers of 2: $2, 4, 8, 16, \ldots$ The 7th term is: 1

(a) 64 **(b)** 14 **(c)** 128 **(d)** 256

2. Which type of angle is the smaller angle between the hands of a clock at 3 o'clock? 1

(a) straight **(b)** acute **(c)** obtuse **(d)** right

3. A Kaprekar step on 6174 gives: 1

(a) 6183 **(b)** 6165 **(c)** 6174 **(d)** 6264

4. On a bar graph with scale 1 cm = 5 units a bar is 6 cm. If its value increases by 20 units, the new height (cm) is: 1

(a) 26 **(b)** 11 **(c)** 10 **(d)** 9

5. The sum of all distinct prime factors of 210 is: 1

(a) 14 **(b)** 20 **(c)** 17 **(d)** 210

6. In the pattern $9, 15, 21, 27, \ldots$ each term is 6 more than the one before. The 15th term is: 1

(a) 93 **(b)** 90 **(c)** 99 **(d)** 105

7. Two angles of 40° and 65° are placed adjacent to each other. The angle left to complete a straight angle is: 1

(a) 75 **(b)** 105 **(c)** 140 **(d)** 255

8. The greatest 4-digit number whose digit sum is 10 is: 1

(a) 5500 **(b)** 9100 **(c)** 9010 **(d)** 8200

9. The total of the two largest values in the table is: 1

Item	Number
Banana	60
Orange	75
Guava	25

(a) 100 **(b)** 135 **(c)** 140 **(d)** 15

10. The number of factors of 54 is: 1

(a) 4 **(b)** 3 **(c)** 8 **(d)** 10

11. In the pattern $7, 11, 15, 19, \ldots$ which term number equals 103? 1

(a) 29 **(b)** 25 **(c)** 26 **(d)** 24

12. What is the angle between the hands of a clock at 5 o'clock, measured the long way round (the larger angle)? 1

(a) 240 **(b)** 180 **(c)** 210 **(d)** 150

13. The smallest EVEN number that can be formed by using each of the digits 5, 6, 7, 2 once is: 1

(a) 5276 **(b)** 2576 **(c)** 7652 **(d)** 2756

14. A pictograph has one symbol = 25. The number of symbols for A is 8 and for B is 3. How many more does A have than B? 1

(a) 125 **(b)** 275 **(c)** 5 **(d)** 150

15. The HCF of 45 and 75 is: 1

(a) 45 **(b)** 15 **(c)** 30 **(d)** 225

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum of the first 30 odd numbers and the first 30 even numbers. 2

17. How much does the hour hand turn from 1:00 to 5:00? How much does the minute hand turn in that time? 2

18. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 2, 3, 1, 4. Find the top number. 2

19. Prepare a frequency table for the data 6, 5, 1, 3, 6, 6, 1, 2, 5, 6, 5, 2, 6, 3 and say which value is repeated the most. 2

20. Find the missing digit so that 456_ is divisible by 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Two ropes of length 48 m and 72 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

22. A row of triangles is made with sticks: 3 sticks for the first triangle and 2 more for each new triangle. Find the sticks needed for 15 triangles, and the number of triangles that can be made with 41 sticks. 3

23. A ray OA is turned about O through a full turn clockwise and then through a quarter turn anticlockwise. Find the net turn in degrees and its type. 3

24. Perform two Kaprekar steps on 2863. 3

OR

Find the fraction of students liking Tennis and the number who do not like it. 3

Sport	Students
Tennis	50
Cricket	20
Hockey	25

25. Find the prime factorisation of 200 and the number of its factors. 3

OR

Find the sum of all odd numbers from 1 to 79. Hence find the sum of all even numbers from 2 to 80. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A school survey data is given. Find the total, the sport with maximum students, and the fraction who play it. Then choose a scale for a bar graph and give the height of the biggest bar. Data: Kabaddi 60, Hockey 20, Tennis 35. **5**

OR

Find the smallest number that leaves a remainder 1 when divided by 4, 5 and 6, and the next such number. **5**

27. A pattern of squares: the first figure has 3 squares and each next figure has 5 more. Find the number of squares in the 12th figure, and the first figure that has more than 108 squares. **5**

OR

A clock shows 2:00. The hour hand turns clockwise until the clock shows 7:00. Find the angle turned by the hour hand, the angle turned by the minute hand in this time, and the smaller angle between the hands at 7:00. **5**

END OF QUESTION PAPER — Total: 50 Marks

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